

Ian Jordy Lopez Diaz

Data Scientist

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Professional Summary

Data Scientist with a Ph.D. in Physics (Statistical Mechanics) and strong experience applying machine learning, statistical modeling, and numerical methods to real-world industrial problems. Currently working at AQTech Power Prognostics, developing data-driven solutions for the energy sector, including predictive modeling, signal analysis, and scalable data pipelines.

Technical Skills

Programming	Python, SQL, R, C/C++
Data Science	Pandas, NumPy, SciPy, Scikit-Learn, TensorFlow, PyTorch, Statsmodels, ETL Pipelines, Data Cleaning, Feature Engineering
Databases	PostgreSQL, MySQL, SQLite
Visualization	Matplotlib, Plotly, Seaborn
Tools	Git, Linux, Jupyter, Docker
Scientific Computing	Monte Carlo Simulations, Numerical Methods, Statistical Modeling

Experience

- 2026–Present **Data Scientist**, *AQTech Power Prognostics*, Florianópolis, Brazil
 - Develop machine learning models for predictive maintenance and fault detection in energy systems.
- 2021–2024 **Data Scientist/Data Analyst/Developer**, *Lima Brain Technology*, Florianópolis, Brazil
 - Designed and implemented data pipelines and forecasting models.
 - Developed Monte Carlo simulations and numerical algorithms.
- 2008–2025 **Physics Lecturer**
 - UFSC, Florianópolis, Brazil; UDESC, Ibirama, Brazil; UFFS, Cerro Largo, Brazil

Education

- 2015–2017 **Ph.D. in Physics (Statistical Mechanics)**, *Federal University of Santa Catarina (UFSC)*, Florianópolis, Brazil
- 2007–2009 **M.Sc. in Physics (Statistical Mechanics)**, *Federal University of Santa Catarina (UFSC)*, Florianópolis, Brazil
- 2003–2006 **B.Sc. in Physics**, *Federal University of Santa Catarina (UFSC)*, Florianópolis, Brazil

Peer-Reviewed Publications

Author of 5 peer-reviewed papers in Statistical Physics and Monte Carlo simulations, published in journals such as *Physica A*, *Physica B*, and *Physical Review E*.